

		Allow other method.
5(a)	Progressive: There is a net transmission of energy from the wave source to points away from the source. Transverse: The direction of the oscillation of the particles in the wave is perpendicular to that of the wave propagation.	[1] [1]
5(b)(i)	intensity $I = \frac{P}{4\pi r^2}$ $= \frac{30}{4\pi(3.0)^2}$ $= 0.265$ $\approx 0.27 \text{ W m}^{-2}$	[1] substitution [1] answer
5(b)(ii)	intensity $I \propto (\text{amplitude } A)^2 \propto \frac{1}{r^2}$ $A \propto \frac{1}{r}$ $\frac{A_2}{A_3} = \frac{r_3}{r_2}$ $= \frac{3.0}{2.0}$ $= 1.5$	[1] substitution [1] answer in decimal
5(b)(iii)	The concave mirror redirects the light, such that all the <u>energy from the bulb is concentrated onto a smaller area</u>. This will lead to a light from the torchlight having a <u>larger intensity</u> than that in (b)(i).	[1] [1]

5

6(a)	+50 V	[1]